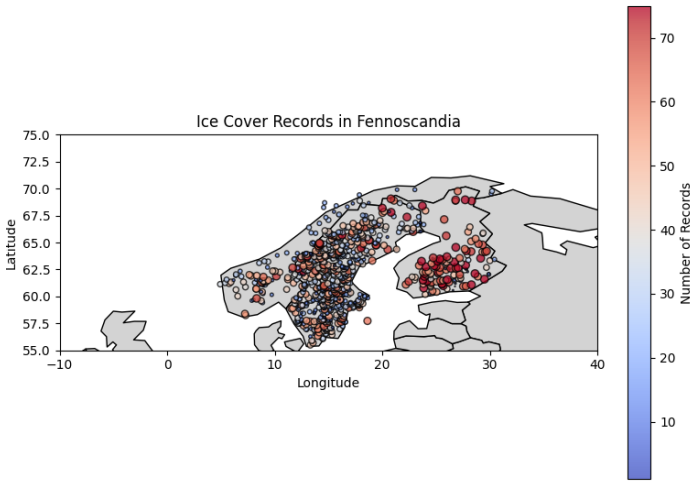
EDA - report on ltbl_ice

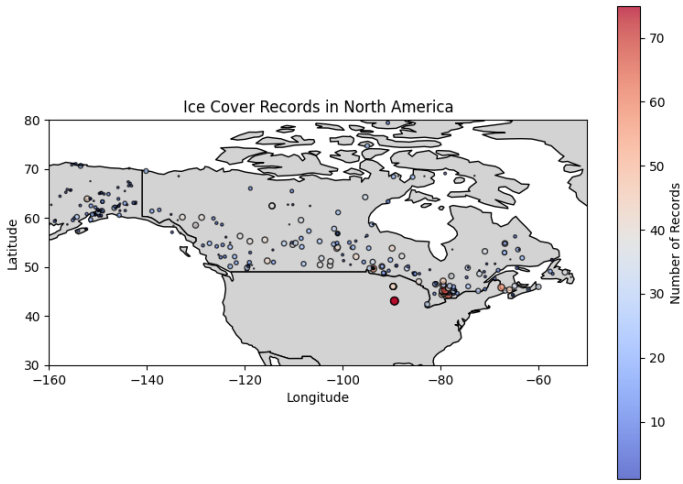
Again, these are some Eda findings about the data in the ltbl_ice table. These are the data about our lakes, in which country they are, their position (latitude, longitude), altitude, and so on …

First I looked at the distribution according to countries:

SE 679
CA 260
FI 121
US 117
NO 102

The number of measurements in data_ip for a lake can be see in this map:





So most of the lakes are in Sweden.

Then here are the missing values:

Column: lake_id
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%

Column: station_id
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%

Column: country
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%

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Column: subset
  Missing Values: 902
  Total Records: 1279
  Percentage Missing: 70.52%
----------------------------------------
Column: lake_name
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%
----------------------------------------
Column: lat_wgs84
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%
----------------------------------------
Column: lon_wgs84
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%
----------------------------------------
Column: altitude_m
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%
----------------------------------------
Column: area_ha
  Missing Values: 0
  Total Records: 1279
  Percentage Missing: 0.00%
----------------------------------------
Column: depth_mean_m
  Missing Values: 746
  Total Records: 1279
  Percentage Missing: 58.33%
----------------------------------------
Column: depth_max_m
  Missing Values: 441
  Total Records: 1279
  Percentage Missing: 34.48%
----------------------------------------
Column: cent_lat_wgs84
  Missing Values: 479
  Total Records: 1279
  Percentage Missing: 37.45%
----------------------------------------
Column: cent_lon_wgs84
  Missing Values: 479
  Total Records: 1279
  Percentage Missing: 37.45%
————————————————————————————————
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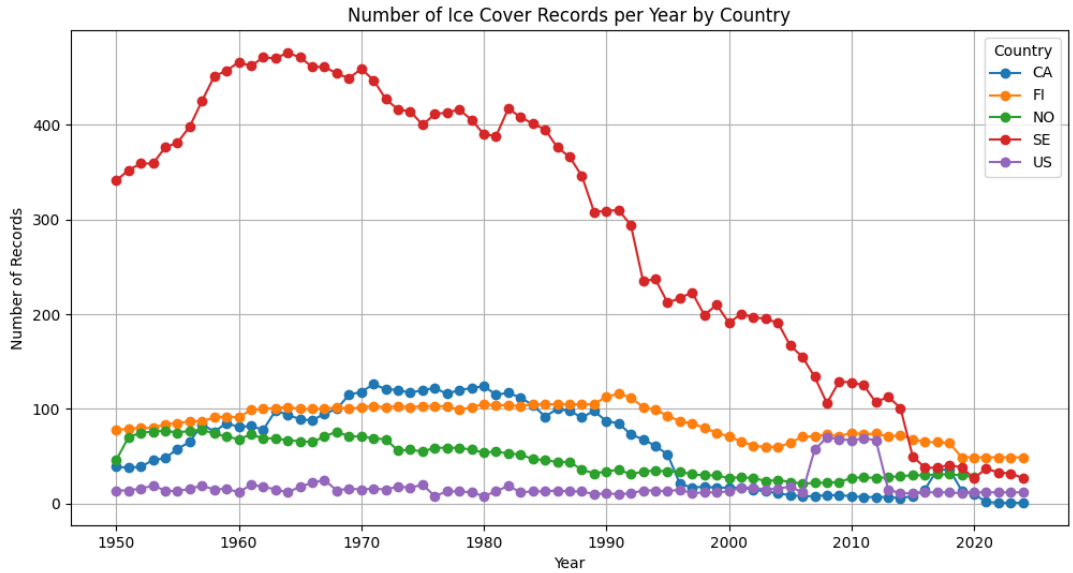
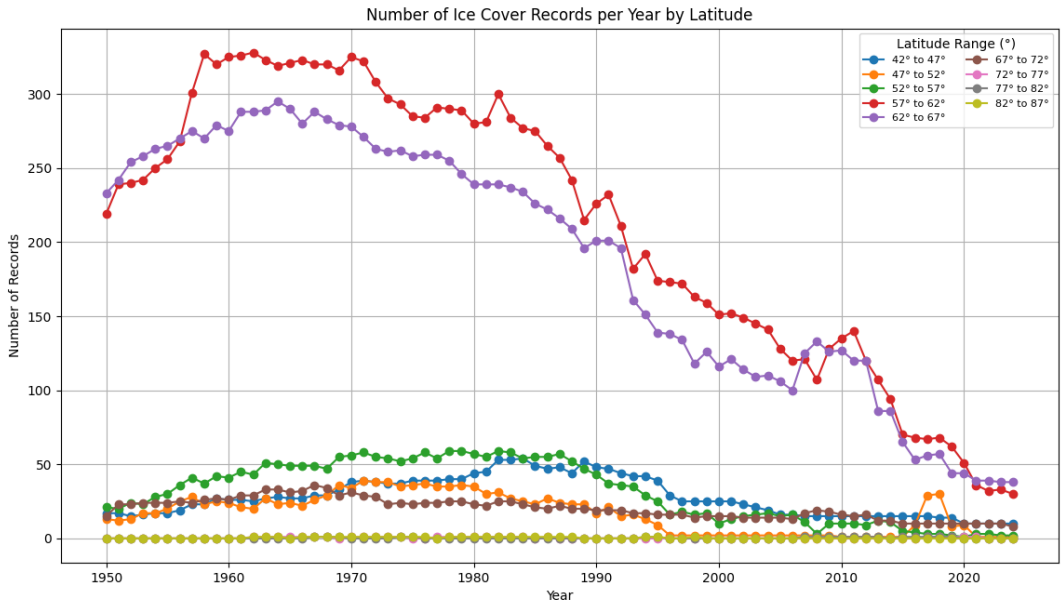
Subset is missing a lot, depth_m and depth_max are also missing quite significantly, then I think that we can maybe ignore those two last columns those are the latitude and longitude, very similar to that lat_wgs84 and lon_wgs84. One are probably coordinates of the lake and another of the station, we have those only for lakes in Sweden. II hope that we dont make much error if we just ignore it.

Then I looked at how much the number of records changed in time in individual countries. It’s in the following graph.

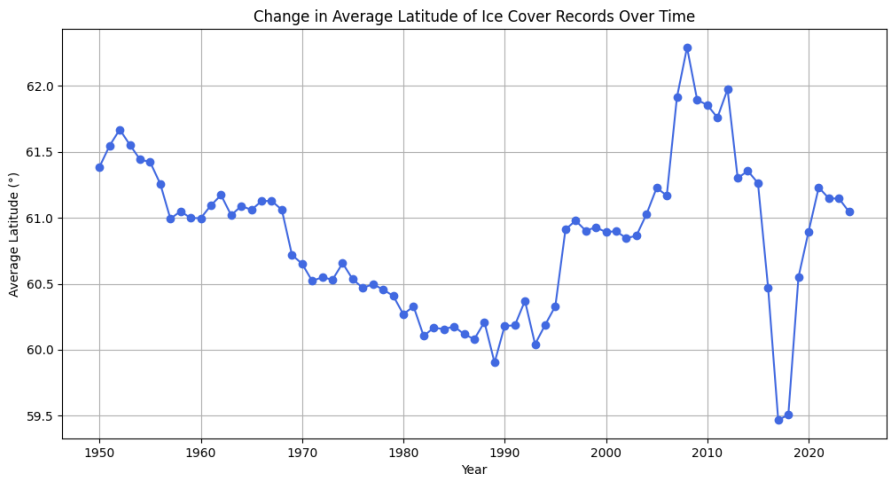


It's quite interesting that there is a significant decline in the number of measurements in Sweden. The number of measurements from Finland is probably the most stable.

Then I checked how changed the number of observations given latitude.



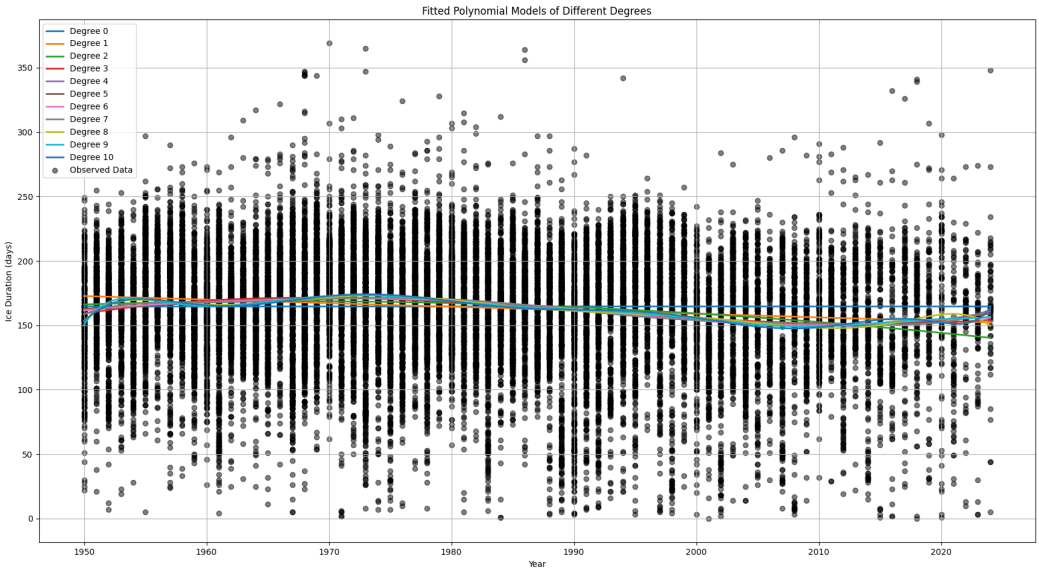
It's hard to say from the graph, whether there has been some non symmetric decline. It would be perhaps quite bad if the recent data were mainly from lower latitudes. Then I plotted the average latitude in a year



The average latitude looks relatively stable so that's a good thing.

I have also created an animation, where one can see for each year where were the measurements collected.

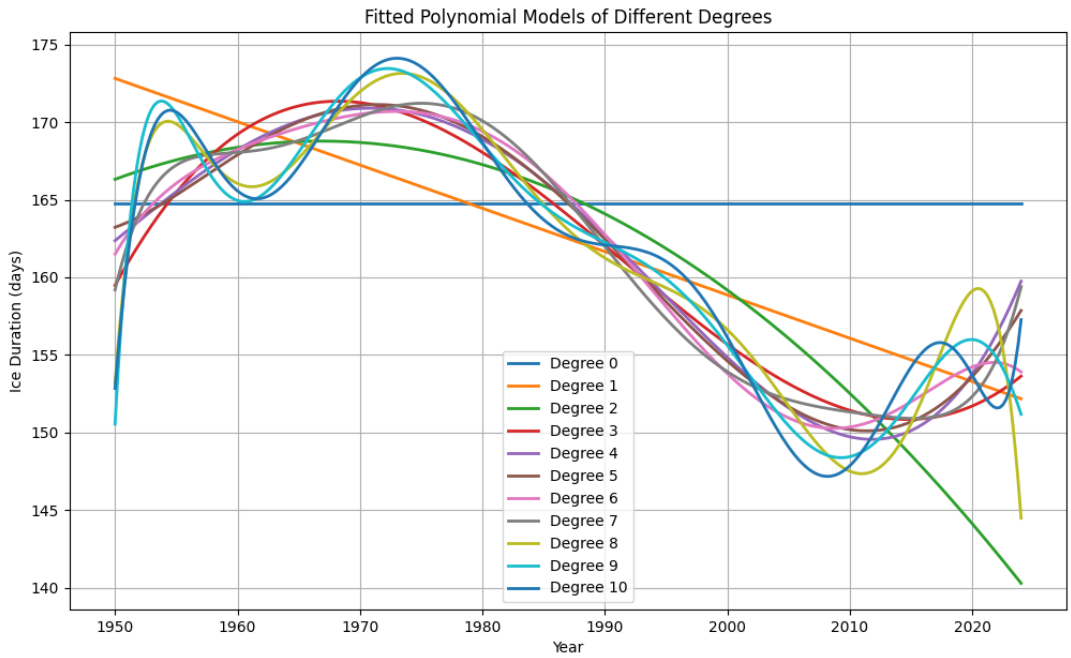
Further I tried to do some more polynomial fitting, also calculating cross-validation so we might have some baseline. I don't know whether it's interesting at all, but I tried to fit polynomials to the



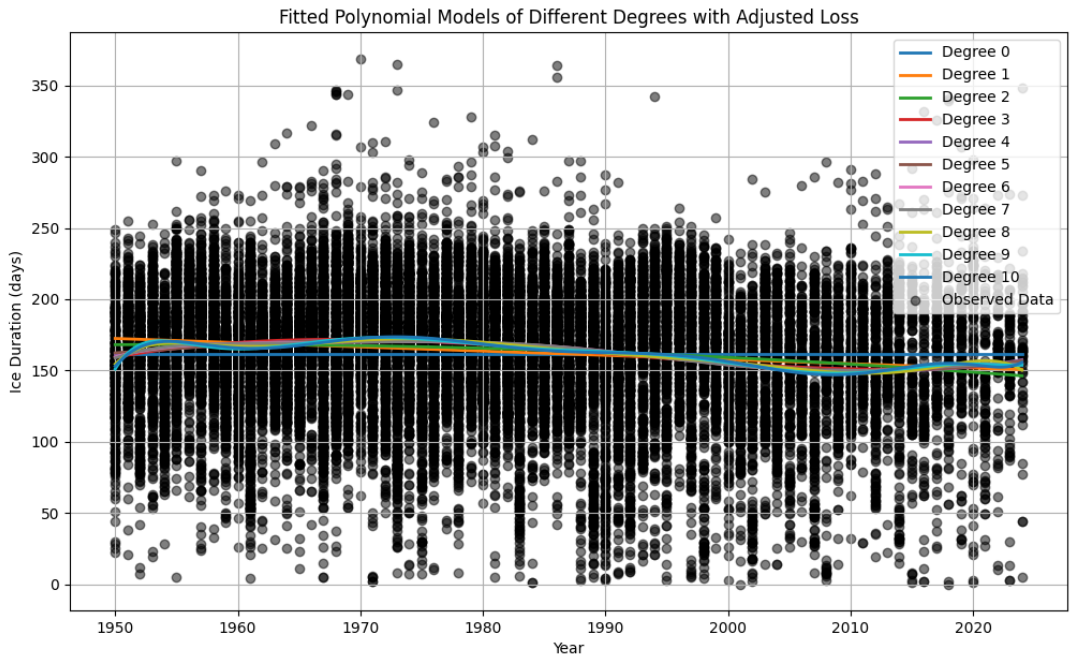
order 10. Here is the fit in the data. Then here is a table with cross-validations (10 splits) mse loss and coefficients:

Degree	CV Loss	Intercept	scaled_year	scaled_year^2	scaled_year^3	scaled_year^4	scaled_year^5	scaled_year^6	scaled_year^7	scaled_year^8	scaled_year^9	scaled_year^10
0.0	1753.8384829821268	164.75367623242857	0.0	nan	nan	nan	nan	nan	nan	nan	nan	nan
1.0	1729.8930003471025	162.5160264161306	-10.213578998082784	nan	nan	nan	nan	nan	nan	nan	nan	nan
2.0	1720.781859149584	165.25619078466772	-12.938210615331444	-11.998701788446239	nan	nan	nan	nan	nan	nan	nan	nan
3.0	1713.2755445414955	163.97773095685557	-23.867835685576193	-7.291266607779362	21.427213362213735	nan	nan	nan	nan	nan	nan	nan
4.0	1712.2952545044755	164.91547726012246	-25.843209539348642	-19.587624008893673	25.102478218528898	16.044252650362175	nan	nan	nan	nan	nan	nan
5.0	1712.400229610906	164.80125341839698	-27.301463710782226	-18.310270131226304	32.74369414099584	14.448034697906543	-7.33854293847318	nan	nan	nan	nan	nan
6.0	1712.2306875955724	165.23052470277503	-28.679880781436815	-29.636889112252632	39.54780996430747	51.63534375012525	-13.692882667932542	-28.654961227138497	nan	nan	nan	nan
7.0	1711.4316601159032	164.7937503006994	-35.96595175887933	-19.50814922528064	112.62923681000402	19.635815263295306	-185.0495262860615	-4.466859396198727	110.19155063760121	nan	nan	nan
8.0	1708.0853221190623	163.16900402344007	-29.092383688370735	51.596664037331266	47.953922804498596	-404.35145178570855	-37.71709666834271	768.1423187047965	16.90224394518181	-427.0996977853157	nan	nan
9.0	1706.9582084110807	163.66794406350974	-18.80292912536177	32.12958519828038	-118.84368494124632	-293.47216647161775	651.9097847634329	571.1674174401566	-1006.737893307688	-319.86058270289925	495.62344904692935	nan
10.0	1706.0557345667708	162.7639824426076	-13.944690884664762	90.94736005141667	-192.7463867536741	-841.3791734619138	947.1816409759147	2290.7227437545557	-1435.5002910665091	-2471.5430312703315	700.1666530274229	926.6485453230491

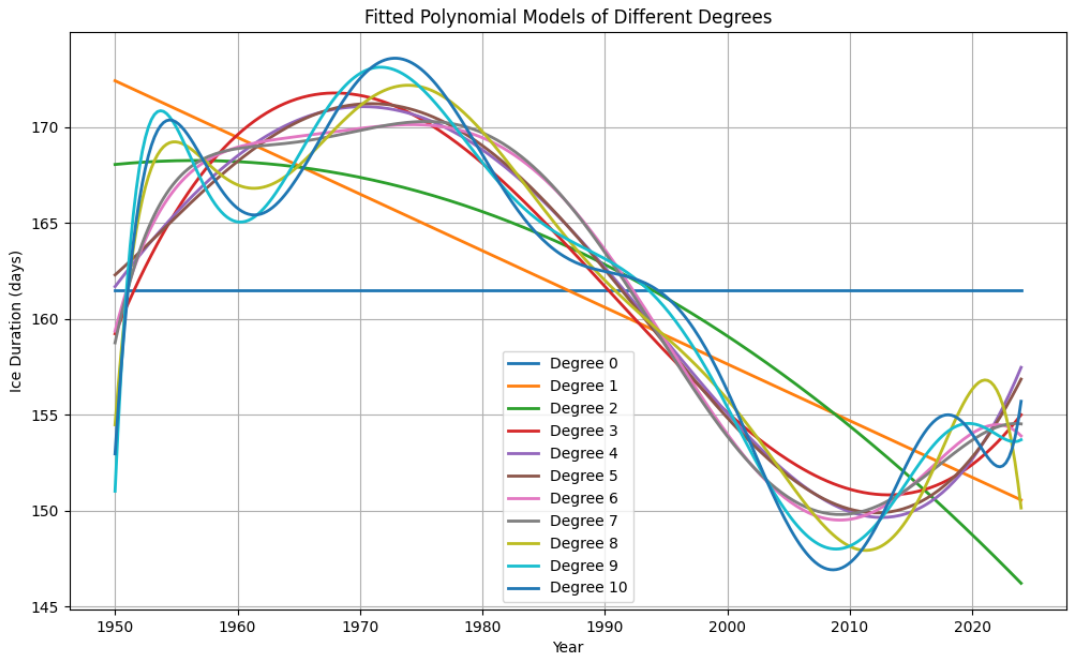
And here are the polynomials by themselves:



And I have tried the similar thing with mse loss that weighted different measurements differently so that the more recent measurements have more weight since there are fewer of them.



Degree	CV Loss	Intercept	scaled_year	scaled_year^2	scaled_year^3	scaled_year^4	scaled_year^5	scaled_year^6	scaled_year^7	scaled_year^8	scaled_year^9	scaled_year^10
0.0	1940.871054649464	161.48032538823952	0.0	nan	nan	nan	nan	nan	nan	nan	nan	nan
1.0	1902.2422693611848	161.48954195718738	-10.928923515600783	nan	nan	nan	nan	nan	nan	nan	nan	nan
2.0	1899.042757567076	163.76646818800796	-10.917828207670428	-6.631485033404346	nan	nan	nan	nan	nan	nan	nan	nan
3.0	1884.4175607355796	163.76535893736448	-25.082551832442906	-6.6512706148710405	22.97159948035032	nan	nan	nan	nan	nan	nan	nan
4.0	1883.5648328233158	164.81746831241838	-25.092357248810575	-16.915637280742622	22.984844191623086	11.676747298700565	nan	nan	nan	nan	nan	nan
5.0	1884.0418767553924	164.8178611905711	-26.461125506495346	-16.91677677695656	29.213485995558	11.675672677057117	-5.46969906493662	nan	nan	nan	nan	nan
6.0	1884.0240289402027	166.021801859698	-26.47478908120634	-41.59869770310345	29.24591770286428	83.90473335458151	-5.504483269172005	-51.694446654272895	nan	nan	nan	nan
7.0	1884.0983627257926	166.021782867043	-28.39066346207156	-41.58540446995494	46.09492487051544	83.86450059423532	-41.7053659968506	-51.66154217419679	21.887510980403757	nan	nan	nan
8.0	1883.2933161145975	164.13949638687055	-28.383322709744867	24.609287509049757	45.978393997721646	-271.8396280372786	-41.341376038915634	550.7799852082084	21.576699950823436	-315.37815132792053	nan	nan
9.0	1880.5491572513922	164.1350899238143	-13.048595781755576	24.671883868107383	-174.04999084087112	-272.24087740273546	797.666590472599	551.3963238720719	-1150.31566279746	-315.6033110073382	541.0880657476824	nan
10.0	1879.6018073060761	163.13983985922889	-13.047622779245078	78.29860371490878	-174.07063966533826	-727.579156620481	797.5053822625736	1889.0505684411615	-1149.9181285637535	-1905.8182776890228	540.900808572865	657.2424471841463



The loss choice has some effect, we can see that the minimum in recent years its lower for the non weighted loss. It is perhaps a bit interesting that as we increase the polynomial degree then the cv loss decreases, I would maybe expect that it starts to overfit a bit, but that is probably because of the large amount of data, like 30000 points. I didn't expect that there would be such an upward trend in recent years in the higher degree models, but that might be of course due to other variables.